

Experiment # 05

Implementation of Gray Level Transformations

Objective

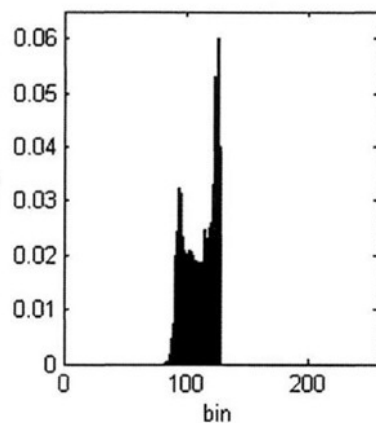
1. To implement contrast stretching.
2. To implement gray level slicing
3. To implement negative, power and log transformation

Software

- Python Idle 3.7 or 3.8.5
- Pycharm/ Spyder

Theory

Contrast stretching is a simple image enhancement technique that attempts to improve the contrast in an image by 'stretching' the range of intensity values it contains to span a desired range of values. Normally (min, max) $\rightarrow$ (0, 255)

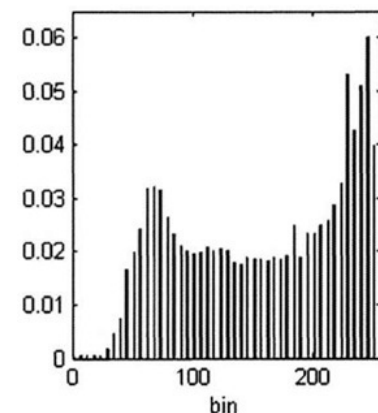
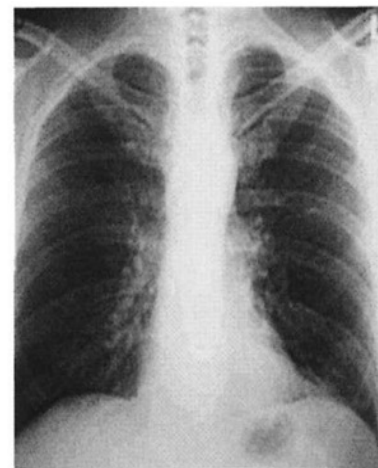


$$p = ((p - \min) / (\max - \min)) * 255$$

Where:

p is pixel value at a certain pixel location

max, min is the max, min pixel value in the entire image



Instead of blindly computing the dynamic range using just the minimum and maximum pixel values, a more robust and adaptive technique is to use the **5th and 95th percentiles** of the input values when deriving the dynamic range of the input image.

The Formula for Contrast stretch or Image Normalization

$$I_o = (I_i - \text{Min}_i) * (((\text{Max}_o - \text{Min}_o) / (\text{Max}_i - \text{Min}_i)) + \text{Min}_o)$$

I_o - Output pixel value

I_i - Input pixel value

Min_i - Minimum pixel value in the input image

Max_i - Maximum pixel value in the input image

Min_o - Minimum pixel value in the output image

Max_o - Maximum pixel value in the output image

Thresholding is the operation through which an image can be converted into a binary image/black & white i.e. having only two distinct levels. Threshold value can be the mean or median etc. of the image.

Gray level Slicing is used to highlight a specific range of gray levels in an image

Transformation operations help enhance the quality of the image by applying operations like log, inverse log and power on the entire image. Different type of transformation yields different results.

The formula of power or gamma transformation is:

$$s = c * r^{\text{(power)}}$$

The formula of log transformation is:

$$s = c * \log(1 + r)$$

The formula of inverse transformation is:

$$s = (L - 1) - r$$

Some Useful Commands:

1. To calculate the mean of 2D array using NumPy: `my_mean = numpy.mean (my_array)`
2. To calculate min (or max) of an array: `my_min = numpy.amin(my_array)`
3. To calculate the power of an array using NumPy: `gamma_corrected = numpy.array (C * (image / max_gray_level) ** gamma, dtype)`
4. To obtain percentile value. `percentile_array = numpy.percentile(my_array, percentile)`
5. To change data type of array. `my_array = my_array.astype(numpy.uint16)`

Lab Tasks:

1: Apply contrast stretching on the image provided in lab by setting 5th and 95th percentiles of the input values to 0 and 255 respectively.

2: Apply the following transformation on an contrast enhanced image.

	Pixel values range	Output pixel value
a)	Less than mean Greater than mean	0 255
b)	Less than mean Greater than mean	255 0
c)	±20 mean Otherwise	0 255

3: After obtaining contrast stretched images from task 1 .

Apply Power Law transformation for the following values of γ (0.2, 0.5, 1.2 and 1.8) . Make sure to adjust data types accordingly.